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Profit allocation mechanisms for carrier collaboration in pickup and delivery service

Bo Dai [a,b,*], Haoxun Chen [b]

[a] Department of Automation, School of Electronic, Information and Electrical Engineering, Shanghai Jiao Tong University, No. 800 Dongchuan Rd., Minhang, Shanghai 200240, PR China
[b] Laboratoire d'optimization des systèmes industriels (LOSI), Institut Charles Delaunay (ICD) and UMR CNRS STMR 6279, Université de Technologie de Troyes, 12 rue Marie Curie, BP 2060, 10010 Troyes Cedex, France

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ABSTRACT

In collaborative logistics, multiple carriers may form an alliance by sharing their transportation requests and vehicle capacities in order to increase vehicle utilization rates and reduce empty back hauls. In this paper, a carrier collaboration problem in pickup and delivery service (CCPPD) is studied. CCPPD has two important issues which are the optimal reallocation of transportation requests among the carriers to maximize a total profit of the alliance and a fair allocation of the profit among the carriers so as to maintain the stability of the alliance. In this study, the profit allocation among carriers is addressed under a centralized collaboration framework. We propose three profit allocation mechanisms which are based on Shapley value, the proportional allocation concept, and the contribution of each carrier in offering and serving requests, respectively. All the mechanisms ensure that the allocation is in the core if the latter is not empty. They are evaluated and compared with numerical experiments on randomly generated instances.

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1. Introduction

In collaborative logistics, multiple carriers may form an alliance by sharing vehicle capacities and transportation requests in order to optimize their transportation operations. The objective of the collaboration is to eliminate empty back hauls, to raise vehicle utilization rates, and thus to increase the profit of each carrier involved. Driven by cost reduction requirement due to fierce competition in the transportation market and increased environmental concerns, improving operational efficiency through resource sharing and collaborative transportation planning is becoming a new business model for transportation industry (D'Amours & Rönnqvist, 2010).

In practice, two approaches may be used to solve collaborative logistics problems: centralized approach and decentralized approach. For centralized approach, Markakis and Saberi (2003) studied a coalitional multicommodity game where players are the nodes of a given network. Each node owns a capacity based on the maximum flow permitted through it. They presented a strategy which first obtains the optimal flow in the network by solving a centralized planning problem and then allocates the total revenue among players in a fair way. Houghtalen, Ergun, and Sokol (2011) studied a network flow problem from a centralized or system perspective, i.e., the networks and demands from all participating carriers are integrated to create a large pseudo-carrier. Audy, D'Amours, and Rönnqvist (2007) proposed different business models for collaborative planning in transportation with an application to the transportation of wood products. They reviewed the decision support systems that use collaborative planning in the wood fiber flow chain in forestry, where a central transportation planner used can be found in Åkar Web (Eriksson & Rönnqvist, 2003) and Virtual Transportation Manager (Audy, D'Amours, & Rönnqvist, 2007).

In addition, Krajewska, Kopfer, Laporte, Ropke, and Zaccour (2008) studied a horizontal cooperation among freight carriers. In the first phase, the profit margins resulting from a central collaborative planning among the carriers are analyzed, whereas the Shapley value is used to determine a fair profit margins allocation among the carriers in the second phase. They demonstrated the effectiveness of their approach by both real-life and artificial instances. Hernandez and Peeta (2011) studied a time-dependent centralized multiple carrier collaboration problem in less-than-truckload industry. They present a collaborative strategy where a central entity (the third party logistics provider, 3PL) is responsible for minimizing the total costs of the carriers. Their experimental results showed the attractiveness of the centralized cooperation paradigm which can significantly increase capacity utilization for carriers and decrease empty-haul trips.

For decentralized approach, Kwon, Lee, and Ma (2005) proposed an integrated multi-round combinatorial auction framework for truckload transportation procurement. In their framework, shippers receive bids on packages of lanes and solve a winner

* Corresponding author at: Department of Automation, School of Electronic, Information and Electrical Engineering, Shanghai Jiao Tong University, No. 800 Dongchuan Rd., Minhang, Shanghai 200240, PR China.

E-mail addresses: Mr.boDai@yahoo.com, Dr.boDai@gmail.com (B. Dai), haoxun.chen@utt.fr (H. Chen).

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determination problem to assign lanes to carriers, whereas carriers solve a bid generation problem (Lee, Kwon, & Ma, 2007) for maximizing their profits and discovering profitable lanes to bid at each round. Price information derived from the solution of the winner determination problem in current round is used by bidders in identifying profitable lanes in next round. Their results indicate that better allocations can be obtained for both shippers and carriers, i.e., shippers reduce costs of procurement of services and carriers are able to identify alternative valuable packages of lanes.

Furthermore, Krajewska and Kopfer (2006) presented a model for the collaboration among independent freight forwarders based on combinatorial auction. Berger and Bierwirth (2010) utilized a decentralized control and auction based exchange mechanism for the reassignment of requests among collaborating carriers. They compared the results of a centralized planning method with those of a decentralized planning method and showed that the cost increase due to the use of the decentralized method is considerable compared with the centralized method. In addition, Dai and Chen (2011b) proposed a price-setting based combinatorial auction framework for CCPPD. Two types of actors involve in the framework, auctioneer and bidders (carriers). The function of the auctioneer is to set and update the price of serving each request, and to maximize the total profit of the whole alliance. Each bidder selects its preferable requests for maximizing its individual profit based on the prices proposed by the auctioneer through solving a bidding problem.

One important issue for collaborative logistics is to design fair profit allocation mechanisms for allocating the total profit among all participants in a collaborative alliance so that the alliance is beneficial to all participants and stable. This issue has gained much attention in recent years. Krajewska and Kopfer (2006) proposed a two steps method to divide the profit gained from the fulfilment of requests among collaborating carriers, which can ensure that the payoff of each carrier is no worse than its payoff before collaboration. In the first step, the minimal execution cost of each request is transferred from the coalition to the carrier who fulfils the request. In the second step, a residual profit is allocated among carriers according to their collaboration advantage indexes defined based on the minimal execution cost of each request, the cost of fulfilling each request by its own carrier (potential self fulfilment cost), and the solution of request reassignment determined by a combinatorial auction. Krajewska et al. (2008) utilised the Shapley value in cooperative game theory to allocate the profit margins generated from the cooperation among multiple carriers. Houghtalen et al. (2011) proposed a profit/cost allocation mechanism based on capacity exchange prices, and they analyzed two approaches for modelling the impact of these prices on the behavior of each individual carrier. Moreover, Houghtalen (2007) introduced a method for implementing the above mechanism, which minimizes the sum of the square distance of each carrier's actual allocation from its target allocation. The target allocation is computed according to some fairness rules, i.e., equal benefits, contribution based on capacity and load, minimum service level. Agarwal and Ergun (2008) investigated a decentralized collaborative multicommodity flow game for carrier collaboration, where individual carriers own capacity on the edges of a network and share the capacity to deliver commodities. A mechanism based on capacity exchange costs and inverse optimization is designed for allocating benefits among carriers, where each carrier follows a strategy to maximize its own profit because it is selfish.

In addition, D'Amours and Rönnqvist (2010) presented a survey of collaborative logistics research. In this survey, profit or cost sharing mechanisms based on weighted costs, separable and non-separable costs, Shapley value, as well the equal profit method are introduced. Xu, Yu, and Zhang (2009) formulated a LTL collaboration game and proposed an allocation method named weighted relative savings model (WRSM), which minimizes the maximum difference between the contribution ratio weighted relative savings among the participants. Computational results for real-life instances showed the effectiveness of WRSM. Frisk, Göthe-Lundgrenc, Jörnstena, and Rönnqvist (2010) considered collaborative transportation planning in a wood supply chain in forestry. They studied a real case in southern Sweden with eight forest companies involved in the collaboration. A number of cost/profit sharing mechanisms based on economic models are investigated, including those based on Shapley value, the nucleolus, separable and non-separable costs, shadow prices and volume weights. An equal profit method with the goal of allocating the participants' relative profits as equal as possible is proposed. The case study proved the effectiveness of their allocation methods. Audy, D'Amours, Lehoux, and Rönnqvist (2010) presented three modes for the collaboration of logistics activities among carriers, vertical mode, horizontal mode, and the combination of vertical and horizontal modes. Five mechanisms for profit sharing are described. The first is based on a predefined incentive rule such as pricing agreements or quantity discount. The second is based on the Shapley value, the nucleolus or the separable and non-separable costs. The third adds an additional constraint related to the profit sharing to a global optimization problem which achieves maximum cost savings. The fourth addresses the global optimization and the profit sharing simultaneously. The fifth partially solves the global optimization problem or its relaxation and provides a partial plan to each business unit. Audy, D'Amours, and Rousseau (2011) addressed some key issues of cost savings sharing in furniture industry, where heterogeneous requirements of collaborating companies may impact their cost savings. They proposed a new cost allocation method which involves two models, i.e., a modified equal profit model and a modified alternative cost avoided model. The method was validated through a case study.

In this paper, we study CCPPD, a carrier collaboration problem appeared in LTL transportation, where each carrier has a set of pickup and delivery requests with time window constraints. It is assumed that an optimal reallocation of transportation requests among carriers to maximize the total profit of their alliance is realized by using a centralized planning framework, which can usually generate a higher profit to the alliance than a decentralized approach (Berger & Bierwirth, 2010). Before collaboration, each carrier serves its own transportation requests by using its own transportation resources (vehicles) and makes its own transportation planning, which leads to a pre-collaboration cost/profit for each carrier. During collaboration, the transportation requests served by each carrier and its transportation plan are determined by solving a global transportation planning problem with the requests and resources of all carriers to maximize their total profit, which leads to the globally optimal total profit of the carriers, referred to as their post-collaboration total profit. The global transportation planning problem is formulated as a mixed integer programming model. Under this centralized planning framework, we address the issue of fairly allocating the total profit among carriers by proposing three profit allocation mechanisms. The first one minimizes the difference between the post-collaboration profit and the Shapley profit of each carrier while ensuring the allocation is in the core (see Section 4.1) if the latter is not empty. The second is based on a well-known proportional allocation concept, Kalai–Smorodinsky solution. It minimizes the difference between the allocation ratios of any two carriers subject to the core constraints. The allocation ratio of each carrier is defined by its post-collaboration profit increase divided by its maximal possible profit increase. The third considers the contribution of each carrier in both offering requests and serving requests. The mechanism is designed based on a model which also minimizes the difference between the allocation ratios of any two carriers while keeping the allocation in the core, where the allocation ratio of each carrier is defined as its

allocated profit divided by its contribution measure. All mechanisms are evaluated and compared with numerical experiments on randomly instances.

The differences between our current work and the relevant studies are listed as following. Krajewska et al. (2008) utilized Shapley value for profit/cost allocation directly, but our first mechanism combines Shapley value with the core concept; in addition, we propose also other two mechanisms. Frisk et al. (2010) designed a mechanism called equal profit method to find a stable allocation such that the maximum difference between relative savings of any two carriers is minimized. For each carrier, its relative savings is defined as its stand alone cost minus its allocated cost and then divided by the first cost. The mechanism proposed by Audy, D'Amours, Rousseau, et al. (2011) is based on the mechanism of Frisk. The only difference between the two mechanisms is that the latter takes account of the special context of transportation operations in furniture industry. Both of the mechanisms only consider costs. The mechanism of Xu et al. (2009) is defined through a weighted relative savings model. They multiply the relative savings of each carrier by a contribution ratio in the equal profit method, and the contribution ratio of each carrier is defined as one minus the value of its contribution to the grand coalition (carrier alliance) divided by the sum of the contribution values of all carriers. A contribution of each carrier is defined as the profit of the grand coalition minus the profit of the coalition excluding the carrier. Our third mechanism is different from the above cited three mechanisms by considering the contribution of each carrier in both offering requests and serving requests and by using a different method to evaluate the contribution of each carrier. Differing from the problem studied by Houghtalen (2007) and Houghtalen et al. (2011), we consider carrier collaboration in pick-up and delivery service in this paper. Their mechanism is built on capacity exchange prices, which means that a carrier receives payments in exchange for its capacities used by other carriers, and vehicles are considered as a type of resource in their study. In our third mechanism, we consider both vehicles and requests as resources to measure the contribution of a carrier. The profit allocation mechanism of Krajewska and Kopfer (2006) is based on two parameters, a potential self-fulfilment cost and a minimal execution cost of each request. Since multiples requests may be served on the same route, it is quite difficult to evaluate the two costs for each request. Our third mechanism can avoid this difficulty in evaluating the contribution of each carrier.

The novelty and contribution of our mechanisms can be found in three aspects: (1) Through the combination with the core concept, our mechanisms can provide a stable allocation. (2) Our third mechanism uses a different way to evaluate the contribution of each carrier, which takes account of the contribution of each carrier in both offering requests and serving requests. (3) The three mechanisms provide carriers with more options in choosing their profit allocation mechanism in collaboration.

The rest of this paper is organized as follows. Sections 2 and 3 describes respectively the carrier collaboration problem studied and outlines a centralized collaboration framework used to address the problem. Section 4 proposes three profit allocation mechanisms and relevant models. Section 5 evaluates the performances of the mechanisms by numerical experiments. Section 6 concludes this paper with remarks for future research. A centralized transportation planning model for the carrier collaboration problem is formulated in the appendix.

2. Problem description

In the CCPPD considered, multiple carriers forms a collaborative alliance in a transportation network to share their transportation requests and vehicle capacities for increasing their vehicle utilization rates and reducing their empty back hauls. Initially, each carrier has acquired certain requests from its customers (shippers) with an ask price for each request, where each request is specified by a pickup location, a delivery location, a time window and a quantity. The ask price of each request is the price paid by a shipper to a carrier for serving the request. This is one type of carrier collaboration problems appeared in LTL transportation. Before collaboration, each carrier makes its transportation planning for serving its acquired requests through its own transportation resources (vehicles), from which the pre-collaboration cost/profit of the carrier can be calculated (the individual profit of each carrier obtained without collaboration). After collaboration, the requests and the transportation resources (vehicles) of all carriers are shared among them, and a centralized collaborative transportation planning is made to maximize a total profit of all carriers, from which the post-collaboration total profit of the carrier alliance can be calculated. Based on the collaborative transportation plan, the total profit is allocated among the carriers by using a profit allocation mechanism. In other word, CCPPD is solved under a centralized collaboration framework in this paper.

The proposed centralized collaboration framework to address CCPPD is realistic in some practical cases. This has been pointed out by other studies on carrier collaboration. One case studied by Krajewska and Kopfer (2006) is a carrier collaboration with a freight forwarder between shippers and carriers. This forwarder acts a third party (non-asset-based) logistics provider that assigns and dispatches shipment requests among multiple asset-based carriers. It is an intermediary who collects transportation requests from the shippers and then assigns the requests to the carriers. The corresponding request assignment problem is solved by the freight forwarder in a centralized way. Another case involves a 4PL (the fourth party logistics provider) in the collaborative transportation planning among carriers (Audy et al., 2007; Bade & Mueller, 1999; Büyüközkana, Feyzioğlua, & Ersoy, 2009;). In 4PL, the carriers entrust an organization to coordinate their transportation activities. The organization can be regarded as a coordinator or a central decision maker. In 4PL, information sharing happens between the carriers and the coordinator but not among the carriers. In this way, the cost and capacity information of each carrier is not revealed to other carriers under the collaboration contract between the carriers and the 4PL, and the confidentiality of these data can be guaranteed.

Furthermore, if all carriers belong to one group company but are independent profit centers, in order to maximize the total profit of the company, the latter may set a central decision maker to coordinate the transportation activities of its carriers. Finally, some decision support systems use a central planner for collaborative transportation planning as that in the wood fiber flow chain in forestry (Audy et al., 2007), Åkar Web (Eriksson & Rönnqvist, 2003), and Virtual Transportation Manager (Audy et al., 2007).

A centralized framework can usually generate more total profit for a carrier alliance than that generated by a decentralized framework (see Berger & Bierwirth, 2010), whereas the decentralized approach provides more autonomy to carriers. The possibility of obtaining more profit by using the centralized framework may attract carriers to accept a centrally coordinated solution if the profit allocation among the carriers is fair. That is why we study the design of profit allocation mechanisms for carrier collaboration in this paper.

From the above discussion, we know there are two important issues that must be addressed for CCPPD. The first issue is a collaborative planning whose objective is to find a set of pickup/delivery requests for each carrier and the vehicle tours for serving the requests that satisfy the vehicle number constraint of each carrier, the capacity constraint of each vehicle and time window

constraints of each request so that the total profit of all carriers is maximized, where each vehicle tour, which is composed of a set of arcs in a transportation network, must start from and return to the depot node of its corresponding carrier. The second issue is to allocate the total profit among all carriers in a fair way so that the post-collaboration profit of each carrier is no less than its pre-collaboration profit and all carriers are willing to remain in the alliance. In this paper, we focus on the second issue.

For simplicity of formulation, we made some assumptions for the CCPPD. Firstly, all requests in the carrier alliance are shareable among its members. Secondly, not all requests must be served by carriers. Thirdly, each carrier has a fixed depot and a limited number of vehicles, and each vehicle must start with and end at its depot. Fourthly, all carriers share a common transportation network but different carriers may have different transportation costs and different vehicle capacities. It should be noted that our profit allocation mechanisms to be proposed are applicable to other carrier collaboration problems without these assumptions. The introduction of the assumptions is only for simplifying the centralized collaborative transportation planning model in the appendix.

To better understand CCPPD described above, we design an illustrative example, which has a transportation network with 21 nodes, three carriers *a*, *b*, *c* with node 5, 17, 11 as their depots, respectively. Carrier *a*, *b*, *c* has 2, 3, 4 vehicles respectively, and each vehicle has the capacity of 10 units. The transportation network is given in Fig. 1. The total number of transportation requests of the carriers is 9, which are represented by *r*1 to *r*9, respectively. For each request, its quantity in units to transport is indicated by a number associated with the arc representing the request. For example, request *r*1 is to pickup 2 units of freight at node 13 and deliver this quantity to node 4. Carrier *a* has requests *r*1, *r*2, *r*3, carrier *b* has requests *r*4, *r*5, *r*6, and carrier *c* has requests *r*7, *r*8, *r*9. The time windows and the price paid by customers for serving each request are given in Table 1. The time window of each depot node is [0,240].

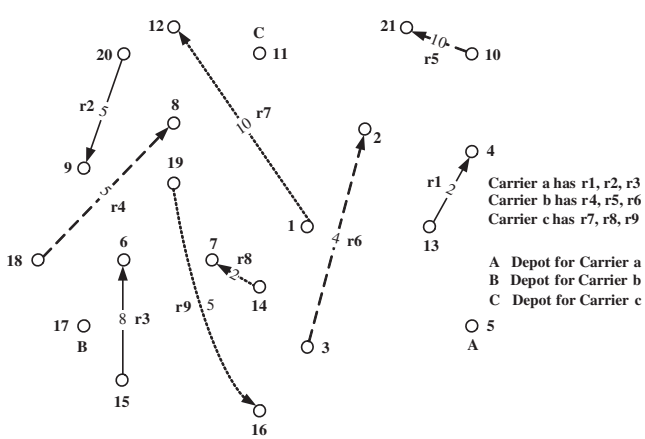

Fig. 1. Transportation network and requests of an illustrative example.

3. Centralized collaboration framework

Two types of actors exist in our proposed centralized collaboration framework for CCPPD, coordinator and carriers. Multiple carriers come to an agreement for constituting a collaborative alliance with a coordinator in charge of making collaborative transportation plans for them. The coordinator is a virtual actor who reallocates the transportation requests of the alliance to all carriers. Its objective is to maximize the total profit of the alliance subject to the constraint that each request is allocated to at most one carrier. In detail, the framework consists of the following steps.

(1) Before collaboration, each carrier calculates its pre-collaboration cost/profit based on its own transportation planning model. In contrast to many other commonly known vehicle routing problems, not all transportation requests have to be executed in the model. Instead, a so-called make-or-buy decision (Goel & Gruhn, 2008) is made for each transportation request to determine whether it should be assigned to a vehicle of the carrier (make) or not (buy).
(2) During collaboration, the coordinator collects the requests of all carriers and put them together. It then allocates a set of requests to each carrier and determines its vehicle tours for serving the requests based on a centralized transportation planning model to maximize the total profit of all carriers, from which the post-collaboration cost/profit of the alliance is calculated. Similarly, a make-or-buy decision is made for each request by the coordinator. The requests allocated to each carrier and its vehicle tours for serving the requests will then be announced to the carrier by the coordinator.

The centralized transportation planning model used in this paper for CCPPD is given in the appendix. Note that the profit allocation mechanisms to be proposed in this paper are also applicable to CCPPD with any other model for optimal reallocation of transportation requests among carriers.

(3) After collaboration, the coordinator allocates the post-collaboration total profit among all carriers by using a profit allocation mechanism. In this paper, we propose three mechanisms for the profit allocation.

4. Profit allocation mechanisms for CCPPD

With the collaboration, the carrier alliance can generate a profit called post-collaboration total profit defined by the optimal objective value of the centralized transportation planning model *CTP* of the carrier collaboration problem studied (see the Appendix), this profit should be fairly allocated among all carriers in the alliance. We design three mechanisms for a profit allocation in this section. The notation to be used in all models defining these mechanisms is given as follows.

Table 1
Time windows, quantities, and service prices of the requests.

Request number	1	2	3	4	5	6	7	8	9
Pickup node	13	20	15	18	10	3	1	14	19
Delivery node	4	9	6	8	21	2	12	7	16
Pickup time window	[109, 130]	[161, 183]	[87, 127]	[89, 122]	[131, 165]	[59, 99]	[97, 128]	[127, 150]	[103, 134]
Delivery time window	[155, 186]	[179, 208]	[179, 206]	[110, 152]	[175, 206]	[183, 210]	[147, 187]	[187, 211]	[115, 152]
Quantity	2	5	8	5	10	4	10	2	5
Service price	29	175	229	75	284	88	173	36	140

Indices	
i, j	node index, $i, j = 1, \ldots, N$, where N represent the number of nodes in the transportation network
m, n	carrier index, $m, n = 1, \ldots, M$, where M represent the number of carriers
k	vehicle index, $k \in V, V = V_1 \cup V_2, \ldots, \cup V_m$, where V_m represents the set of vehicles owned by carrier m
l	request index, $l = 1, \ldots, L$, where L represents the number of requests of all carriers
Parameters	
CS	the carrier set, which includes all carriers in the alliance
r'_m	pre-collaboration profit of carrier m before collaboration
$r(CS)$:	post-collaboration total profit of the carrier alliance after collaboration
R_m	total revenue of the requests offered by carrier m and served after collaboration
$r(S)$	post-collaboration total profit of carrier coalition S, $S \subseteq CS$
$c(S)$	post-collaboration total cost of carrier coalition S, $S \subseteq CS$
c_m	post-collaboration cost of carrier m after collaboration
Variables	
r_m	real decision variable, which represents the profit of carrier m after collaboration and is referred to as post-collaboration profit of the carrier hereafter
f	real auxiliary variable, which represents the maximum difference between the profits or allocation ratios of any two carriers and is taken as the objective function of each mechanism to minimize

Note that, r_m and f are the only decision variables appeared in each of three mechanism design models to be formulated later. Hence, all these models are linear programming models.

4.1. Principles of core guaranteed mechanisms

For profit allocation among players in a cooperative game, some well known solution concepts have been proposed, such as Core concept (Gillies, 1959), Shapley value (Shapley, 1953), Kalai–Smorodinsky solution (Kalai, 1977). The core is the set of imputations (Kalai, 2008) under which no coalition has a value greater than the sum of its members' payoffs. If an imputation is in the core, no coalition has incentive to leave the grand coalition (carrier alliance) and receive a larger payoff. The core allocations are very important since only such allocations are stable. Unfortunately, neither the allocation based on Shapley value nor that of Kalai–Smorodinsky can ensure its inclusion in the core. As a result, the two allocations may be not stable. For this reason, in this study, we try to design mechanisms of stable allocation by combining Shapley value, Kalai–Smorodinsky solution and other solutions with the core concept. Our objective is to design profit allocation mechanisms that are fair in some sense and can ensure the sustainability of the grand coalition in carrier collaboration. Three mechanisms will be proposed. The first one combines Shapley value with the core concept. The second combines Kalai–Smorodinsky solution and the core concept. The third is based on the contribution of each carrier to the alliance in both offering and serving requests while guaranteeing the allocation is also in the core.

4.2. Core guaranteed Shapley allocation mechanism

Shapley value (Shapley, 1953) is a wide-accepted profit allocation mechanism in cooperative games. It is derived based on the axioms of efficiency, symmetry, dummy, and additivity, however it may be not in the core and thus can not assure the stability of the grand coalition. A *Core guaranteed Shapley allocation mechanism* (*SAM*) we propose can be formulated by a linear programming model, denoted by *P1* and given by Eqs. (2)–(7), where r''_m is a parameter defined by Eq. (1), which denotes the post-collaboration profit of carrier m given by the Shapley value.

$$r''_m = \sum_{S \subseteq M \setminus \{m\}} \frac{|S|!(M - |S| - 1)!}{M!} \cdot (r(S \cup \{m\}) - r(S)),$$
$$m = 1, \ldots, M, \tag{1}$$

Model *P1*:

$$Z^l = Minf \tag{2}$$

Subject to:

$$r_m - r''_m \leqslant f, \quad \forall m = 1, \ldots, M, \tag{3}$$

$$r_m \geqslant r'_m, \quad m = 1, \ldots, M, \tag{4}$$

$$\sum_{m \in S} r_m \geqslant r(S), \quad \forall S \subset CS \tag{5}$$

$$\sum_{m=1}^{M} r_m = r(CS) \tag{6}$$

$$r_m, f \in \Re, \quad m = 1, \ldots, M, \tag{7}$$

The model is to find a profit allocation which minimizes the difference between the post-collaboration profit and the Shapley profit of each carrier, while keeping the allocation in the core. Constraints (4) assure the individual rationality of the allocation. Constraints (5) ensure the stability of the grand coalition. Constraint (6) implies that the allocation is efficient (budget-balanced). Constraints (4)–(6) together ensure that the allocation is in the core.

Three situations may happen for model *P1*:

(1) The allocation is in the core and the same as the Shapley allocation.
(2) The allocation is in the core but different from the Shapley allocation.
(3) The core is empty, the allocation does not exist.

Compared with the classic Shapley allocation that may be not stable, this mechanism can generate a stable allocation if the core exists. The objective of *SAM* is to find a feasible allocation that stays in the core and is as close to the Shapley value as possible.

4.3. Core guaranteed proportional allocation mechanism

Kalai–Smorodinsky (KS for short) solution (Kalai, 1977) is a well-known concept in bargaining games. It has the properties of symmetry, Pareto optimal, and individually monotonic. For a bargaining game with two players, suppose that player 1 could receive a gain d_1 without cooperation of player 2 and could receive a maximum gain I_1 through the cooperation of player 2; vice versa, suppose that player 2 could receive a gain d_2 alone and could receive a maximum gain I_2 through the cooperation of player 1. The KS solution is the solution (u_1, u_2) satisfying the condition $(u_1 - d_1)/(I_1 - d_1) = (u_2 - d_2)/(I_2 - d_2)$, where u_1 and u_2 denote the actual gain of player 1 and player 2, respectively. Kalai (1983) generalized

the solution concept to n-player bargaining games, with the number of players fixed to n, i.e., the games for which the cooperation only happens in the grand coalition. The KS solution is then the solution $(u_1, u_2, \ldots, u_n)$ satisfying the condition $(u_i - d_i)/(I_i - d_i) = (u_j - d_j)/(I_j - d_j)$ for any $i, j = 1, \ldots, n, i \neq j$. To the best of our knowledge, KS solution was never used in profit allocation for carrier collaboration.

Inspired by the n-player KS bargaining solution, a *Core guaranteed proportional allocation mechanism* (*PAM*) is proposed, which can be formulated by a linear programming model *P2* and given by equations (9) and (10). In *P2*, r'''_m is a parameter which represents the maximal post-collaboration profit that could be given to carrier m by assuming the individual rationality of each carrier, i.e., constraints (4) are satisfied. This parameter may be calculated in several ways, for instance, by Eq. (8). This equation implies that the total profit allocated to the coalition constituted by all carriers excluding carrier m should no less than the total profit that the sub-coalition can generate with the cooperation of its members; otherwise, one of the members may drop out from the grand coalition including carrier m.

$$r'''_m = r(CS) - r(CS \setminus \{m\}), \quad m = 1, \ldots, M, \tag{8}$$

Model *P2*:

$$Z^{II} = Minf \tag{9}$$

Subject to constraints (4)–(7) and:

$$(r_m - r'_m)/(r'''_m - r'_m) - (r_n - r'_n)/(r'''_n - r'_n) \leqslant f, \quad \forall m, n = 1, \ldots, M, \quad m \neq n, \tag{10}$$

The objective of the model is to find a profit allocation which minimizes the difference between the proportional allocation ratios of any two carriers subject to the core constraints ensuring that the allocation is in the core, where the proportional allocation ratio of each carrier is defined as its post-collaboration profit increase divided by its maximal profit increase, i.e., $(r_m - r'_m)/(r'''_m - r'_m)$ for each carrier m. Note that, in Eq. (10), only r_m and r_n are decision variables, r'_m, r'''_m, r'_n, r'''_n are parameters. Therefore, equation (10) is also a linear constraint, which ensures that model *P2* is still a linear programming model.

Three situations may also happen for model *P2*:

(1) The allocation is in the core and the same as the proportional allocation of the KS solution.
(2) The allocation is in the core but different from the proportional allocation of the KS solution.
(3) The core is empty, the allocation is not found.

4.4. Core guaranteed contribution-based allocation mechanism

The two mechanisms proposed in the above are based on Shapley value and Kalai–Smorodinsky proportional solution concept, respectively. Both of them are belong to axiom-based solution concept, i.e., a unique allocation is derived by first making some axiomatic assumptions, and if all players in a cooperative game agree with the assumptions, they should also accept the allocation. However, for the Shapley value and the KS propositional solution, each has one axiomatic assumption that is not universally accepted: the additivity axiom for Shapley value (Banker, 1981) and the monotonicity axiom for KS solution (Felsenthal & Diskin, 1982). Some researchers have proposed other solution concepts by replacing one questionable axiomatic assumption of Shapley value or KS solution with a new (still questionable) axiomatic assumption. In this subsection, we try to propose a new allocation mechanism which is based on some insights on the contribution of each carrier to its coalition rather than based on axiomatic assumptions.

As it is widely recognized, carrier collaboration is realized by sharing transportation resources (vehicles) and requests among carriers. The contribution of each carrier should therefore be evaluated based on the contribution of each carrier in offering and serving requests. Here, offering a request means that a transportation request acquired by a carrier is transferred to another carrier who will service (execute) the request. If a carrier offers (outsources) a request to another carrier, the offering carrier should gain a profit, by retaining part of the revenue of the request, as an award to its effort made in acquiring the request from its client. This part depends on the revenue of the request. At the same time, the offering carrier should transfer part of the revenue to the service carrier so that the latter can also gain a profit by serving the request. The profit is the difference between the revenue transferred by the offering carrier and the cost of serving the request by the service carrier. As the cost can be used to measure the effort made by the service carrier to serve the request, the profit should depend on this cost.

Unfortunately, for each individual request, it is not easy to determine a fair division of its revenue between its offering carrier and its service carrier for the two reasons: (1) some requests may be more profitable than other requests, and (2) the cost of serving each individual request is not easy to be evaluated since multiple requests may be served on the same route. Therefore, we should take a global view in evaluating the fair profit gained by offering requests and the fair profit gained by serving requests for each carrier.

From the insights obtained from the above discussion, we define parameter w_m, which measures the contribution of carrier m, by Eq. (11), where θ_1 and θ_2 are two positive weights. The numerator of the fraction on the right side of the equation is a weighted sum of the cost of serving all requests assigned to the carrier (c_m) and the revenue of all requests offered by the carrier and served by other carriers (R_m). The denominator of the fraction is a weighted sum of the total cost of serving requests ($c(CS) = \sum_{m=1}^{M} c_m$) and the total revenue of all requests offered and served by carriers ($\sum_{m=1}^{M} R_m$). Obviously, $w_m \in [0, 1]$ and $\sum_{m=1}^{M} w_m = 1$.

$$w_m = (\theta_1 \cdot c_m + \theta_2 \cdot R_m) \Big/ \left(\theta_1 \cdot \sum_{m=1}^{M} c_m + \theta_2 \cdot \sum_{m=1}^{M} R_m \right), \quad m = 1, \ldots, M, \tag{11}$$

For example, given an optimal solution of the centralized planning model *CTP* with three sets of values, x_{ij}^k (number of times that arc (i, j) is visited by vehicle k), y_{lm} ($y_{lm} = 1$ if request l is allocated to carrier m and 0 otherwise), and z_{lm} ($z_{lm} = 1$ if request l is offered by carrier m and 0 otherwise), the parameters c_m and R_m can be calculated respectively as follows, $c_m = \sum_{k \in V_m} \sum_{i=1}^{N} \sum_{j=1, j \neq i}^{N} c_{ij}^m \cdot x_{ij}^k$ and $R_m = \sum_{l=1}^{L} p_l \cdot z_{lm} \cdot \sum_{m=1}^{M} y_{lm}$.

The remaining thing is to determine the two weights θ_1 and θ_2. Intuitively, θ_1 and θ_2 represent respectively the importance (contribution) of serving requests and the importance (contribution) of offering requesting to the creation of the total profit of the coalition. To get an insight into the relative importance of each component (request offering and request service) in creating the profit, we consider a particular case with two carriers in which one carrier has requests but without any vehicle to serve the requests and the other carrier has enough vehicles to serve all requests but without any request. This extreme case, which does not mix request offering with request service at each carrier, permits us to see more clearly the contribution of each component in profit creation. Our question for this case study is: if the two carriers form a collaborative coalition to share requests and vehicle capacities, that is, all

requests of the first carrier will be served by the second carrier, how should the total profit of the coalition be divided between the two carriers?

Suppose that the total revenue of all requests of the first carrier is p and the total cost for serving all requests by the second carrier is c with $p > c$. The total profit of the coalition is thus $p - c$. Certainly, each carrier gets nothing (with zero profit) if they do not cooperate each other (we assume that the first carrier cannot subcontract its requests to a third party and the second carrier cannot get requests from a third party). Then, if we divide the total profit between the two carriers based on the Shapley value, we can get an allocation in which each carrier gets a profit of $(p - c)/2$. The same allocation result can be obtained if we apply the KS proportional solution concept to divide the total profit. For this allocation, the first carrier creates the profit $(p - c)/2$ by offering requests whose total revenue is p, so the profit created by per unit of the revenue is $(p - c)/2p$; the second carrier creates the profit $(p - c)/2$ by serving requests with total service cost c, so the profit created by per unit of the cost is $(p - c)/2c$. Since the contribution of serving requests and the contribution of offering requesting to the total profit are measured by cost and revenue respectively, the weights θ_1 and θ_2 for this example should be proportional to $(p - c)/2c$ and $(p - c)/2p$, that is, proportional to $(p - c)/c$ and $(p - c)/p$.

From the above discussion, we define θ_1 as the total profit of all carriers divided by the total cost of all carriers after collaboration, and define θ_2 as the total profit of all carriers divided by the total revenue of all carriers. θ_1 and θ_2 can be calculated before profit allocation. Economically, θ_1 is called *Profit Margin on Cost* (Shaikh, 1978) and θ_2 is called *Gross profit margin* (Magoon, 2008).

With the contribution measure for each carrier defined above, we propose a *Core guaranteed contribution-based allocation mechanism* (*CAM*) described by model *P3*, which is a linear programming model and given by Eqs. (12) and (13).

Model *P3*:

$$Z^{III} = Minf \tag{12}$$

Subject to constraints (4)–(7) and:

$$(r_m - r'_m)/w_m - (r_n - r'_n)/w_n \leqslant f, \quad \forall m, \\ n = 1, \ldots, M, \quad m \neq n, \tag{13}$$

The objective of the model is to find a profit allocation which minimizes the difference between the contribution-based allocation ratios of any two carriers subject to the core constraints ensuring that the allocation is in the core, where the contribution-based allocation ratio of each carrier is defined as its post-collaboration profit divided by its contribution measure, i.e., r_m/w_m for each carrier m. The left term of constraint (13) is the difference between the allocation ratios of carrier m and carrier n. Note that with the core constraints (4)–(7), the contribution-based allocation mechanism can ensure personal rationality of each carrier and the stability of the grand coalition.

To avoid the case of division by zero when $w_m = 0$ or $w_n = 0$, we rewrite constraint (13) as (14), which does not change model *P3*.

$$(r_m - r'_m) \cdot w_n - (r_n - r'_n) \cdot w_m \leqslant f \cdot w_m \cdot w_n, \quad \forall m, \\ n = 1, \ldots, M, \quad m \neq n, \tag{14}$$

Furthermore, in order to ensure that a carrier with a larger contribution measure is allocated with a profit no less than that allocated to a carrier with a smaller contribution measure, we may add the following constraints to model *P3*.

$$(r_m - r'_m) \cdot Sign(w_m - w_n) \geqslant (r_n - r'_n) \cdot Sign(w_m - w_n), \quad \forall m, \\ n = 1, \ldots, M, \quad m \neq n, \tag{15}$$

If model *P3* is solved with constraints (15), the obtained allocation is referred to as *contribution-monotone allocation*. Under the condition that the core exists and all constraints (15) are satisfied, the mechanism will allocate more profit to a carrier that has a larger contribution measure.

4.5. Comparison of three mechanisms

Each of the above three mechanisms defines an imputation in cooperative games (Kalai, 2008). The imputation is a feasible allocation with Pareto optimal and individually rational properties for the grand coalition. Shapley value defines an imputation satisfying some axioms and according to the expected marginal contribution of each player to each possible coalition. KS solution is the unique imputation with profit gains proportional to the players' ideal gains. Our proposed contribution-based profit allocation mechanism defines an imputation based on the contribution of each carrier in both offering requests and serving requests. These three mechanisms may lead to different imputations for the same carrier collaboration problem. To show their differences, we consider the following example.

Suppose that three carriers (a, b, c) are involved in the collaboration. The parameters are given as following. The total ask price of the requests of each carrier: $p_a = 0$, $p_b = 100$, $p_c = 100$. The pre-collaboration profit of each carrier: $r'_a = 0$ (carrier a does not have any request to serve before collaboration), $r'_b = 10$, $r'_c = 10$. The post-collaboration cost of each carrier and all carriers: $c_a = 20$, $c_b = 0$, $c_c = 100$, $c(\{a, b, c\}) = 120$. The post-collaboration profit of each coalition: $r(\{a, b\}) = 30$, $r(\{a, c\}) = 25$, $r(\{b, c\}) = 20$, $r(\{a, b, c\}) = 80$.

For this example, three mechanisms generate three different allocations. The post-collaboration profit of each carrier given by *SAM* is $r^S_a = 25.8333$, $r^S_b = 28.3333$, $r^S_c = 25.8334$. The post-collaboration profit of each carrier given by *PAM* is $r^P_a = 24.8276$, $r^P_b = 28.6207$, $r^P_c = 26.5517$. The post-collaboration profit of each carrier given by *CAM* is $r^C_a = 5$, $r^C_b = 25$, $r^C_c = 50$. As we can see, carrier a obtains the biggest profit increase in *SAM* and *PAM* although it only takes charge of 1/6 of the total cost and does not provide any request to other carriers. Thus, the profit allocations given by *SAM* and *PAM* may be not very fair to carrier b and carrier c. However, in *CAM*, carrier a obtains the smallest profit, and carrier c gains the biggest profit because it takes charge of 5/6 of the total cost and provides 1/2 of the total revenue. This allocation seems more reasonable.

If the cost of each carrier is changed to $c_a = 100$, $c_b = 0$, $c_c = 20$, the profit allocations of *SAM* and *PAM* are not varied. However, in *CAM*, the profit of each carrier is changed to $r^C_a = 25$, $r^C_b = 25$, $r^C_c = 30$, and carrier a now obtains the biggest profit increase because it takes charge of 5/6 of the total cost. In this situation, three mechanisms generate similar profit allocations.

It is hard to say which profit allocation mechanism is better than the others. In practice, one of the mechanisms may be chosen based on an agreement of all carriers, who must decide the way of allocating the total profit before their collaboration.

4.6. Case with empty core

The three mechanisms proposed in the above can not be applied when the core is empty. To overcome the difficulty, we can generalize them by introducing a new model *EC* for seeking a γ-budget balanced allocation (Caprara & Letchford, 2010). The value of γ is between 0 and 1, and it is maximized through searching a γ-approximate non empty core.

Model *EC*:

$$Z^{EC} = Max\gamma \tag{16}$$

Subject to constraints 4, 5, 7 and:

$$\sum_{m=1}^{M} r_m \geqslant \gamma \cdot r(CS) \tag{17}$$

$$\gamma \in \Re, \gamma \in (0,1] \tag{18}$$

After obtaining maximum γ by solving model *EC*, we can replace constraint (6) in model *P1*, or *P2*, or *P3* by constraint (17) and then solve the model again, in this way a γ-approximate allocation can always be found.

5. Numeric experiments

In this section, we design some numeric experiments to evaluate our proposed three mechanisms. Up to now, there are no benchmark instances for CCPPD, so twenty randomly generated instances are used to evaluate the performances of the mechanisms. Each instance involves three carriers ($K = 3$). The number of nodes is set to 21 ($N = 21$, 3 nodes as the depot nodes of three carriers, the other 18 nodes as the customer nodes). The coordinates of each node are randomly generated from 42×42 square. After the coordinates of all nodes are generated, the Euclidean distance d_{ij} between any two nodes i and j is calculated. Without loss of generality, we set $d_{ij} = \tau_{ij}^k = c_{ij}^m$ in the centralized transportation planning model (see the Appendix). Every carrier randomly selects a node as its depot node which is different from the depot nodes of all other carriers. Each carrier owns a number of vehicles randomly chosen from 1 and 10, and each vehicle has the same capacity of 10 units.

The number of transportation requests L is set to $(N - K)/2$ ($L = 9$). Each request is associated with 2 nodes, pickup node and delivery node. The requests are generated by randomly choosing a pickup node i ($i \neq$ any depot) and a delivery node j ($j \neq i$ and $j \neq$ any depot), each node is associated with at most one request. Each request is associated with a randomly generated quantity of freight u units which is no larger than a predefined number, which is set to 10. Given a request with pickup node i and delivery node j for carrier k, the ask price paid by a shipper to a carrier for serving the request is defined by $\mu \cdot (1 + \alpha) \cdot \beta \cdot (c_{oj}^m + c_{ij}^m + c_{jo}^m)$ which is set in the same way as that in the paper (Dai & Chen, 2011a), where o denotes the depot node of the carrier and $(c_{oj}^m + c_{ij}^m + c_{jo}^m)$ is the direct shipping cost of the request, β is the average utilization of vehicle in an ideal situation for serving the request, α denotes the minimum profit margin of the carrier, and μ is an adjustable parameter no less than 1. Parameter α is set to 5% for all carriers according to Holguín-Veras, Xu, Jong, and Maurer (2011). Parameter μ is set to 2, which ensures that the ask price of the request is sufficiently larger than its service cost in order to avoid the situation that no carriers serves the request. The time windows are generated in the following way: the time interval for serving all requests is set to [0,240] (480 min = 8 h, time unit is taken as 2 min), the earliest service time a_i at pickup node i is randomly chosen in $[\tau_{oi}, 240 - \tau_{oj} - \tau_{ij}]$, where τ_{oi} is calculated by considering the farthest depot o (three depots in total) from node i, the latest service time b_i is randomly chosen from $(a_i + 15)$ to $(a_i + 45)$, the earliest service time a_j at delivery node j is randomly chosen in $[\tau_{oi} + \tau_{ij}, 240\text{-}\tau_{oj}]$, and latest service time b_j is randomly chosen from $(a_j + 15)$ to $(a_j + 45)$. And the condition $(b_j - a_i) > c_{ij}$, $b_i \leqslant 240$ and $b_j \leqslant 240$ are verified during the time windows generation process. The time window for the depot of each carrier is set to [0,240].

All models involved are solved by using Cplex 12 for each instance. The computational results are given in Tables 2 and 3, where the profit of each carrier before collaboration, the post-collaboration total profit of the alliance, the post-collaboration profit of each carrier given by mechanism 1 (*SAM*), the post-collaboration profit of each carrier given by mechanism 2 (*PAM*), the post-collaboration profit of each carrier given by mechanism 3 (*CAM*) are showed.

As shown in Tables 2 and 3, for each instance, all the three mechanisms can generate a profit allocation which stays in the core and the profit of each carrier never decreases in the collaboration with all other carriers. *SAM* can find a profit allocation which is the same as that given by classic Shapley value (the objective (2) is equal to 0) for all instances except for instances 6, 18 and 19. Meanwhile, *PAM* can generate the same profit allocation as Kalai–Smorodinsky proportional mechanism (the objective (9) is equal to 0) for all instances. Furthermore, *CAM* can yield a *contribution-monotone* allocation for instances 1, 7, 10, 11, 12, 13, 16, and 20, and yield a feasible allocation for all other instances.

From the experimental results, we can see that the profit increases of each carrier after collaboration generated by three mechanisms may be different. This is because the core of each instance contains multiple feasible allocations, and a different definition of fairness in each mechanism gives rise to a different core allocation. For some instances, the profit increase of each carrier is only slightly changed if the profit allocation mechanism used is changed from one to another, whereas for the other instances the profit change is not negligible. *SAM* leads a core allocation which is close to the classic Shapely value, *PAM* creates a core allocation which is close to the proportional Kalai–Smorodinsky solution, and *CAM* generates a core allocation based on the contribution of each carrier to their alliance, which minimizes the maximum difference between the contribution-based allocation ratios of any two carriers. The fairness of profit allocation is interpreted in different ways in the three mechanisms. The fairness of *SAM* is represented by allocating the profit according to the weighted average of the contribution of each carrier to each possible coalition. The fairness of *PAM* is reflected by assigning the profit proportionally to carriers, where the allocation ratio of each carrier is determined by the maximum profit that this carrier can obtain with collaboration and the disagreement profit that the carrier can obtain without collaboration. The fairness of *CAM* is expressed in distributing the profit in line with the contribution of each carrier to the alliance, which considers both offering requests and serving requests by each carrier.

Since no universally accepted fairness definition exists for profit allocation in cooperative games, which mechanism is chosen for a practical carrier collaboration application should be determined by the common preference of all carriers involved in the collaboration. We cannot say one mechanism is better than another, but the proposed three mechanisms provide carriers with more options in choosing their profit allocation mechanism

6. Conclusion

A carrier collaboration problem with pickup and delivery requests and time window constraints has been studied in this paper. Three profit allocation mechanisms and relevant models are proposed under a centralized collaboration framework. Numerical experiments on randomly generated instances are used to evaluate and compare the mechanisms. Our future work is to find practical applications of these mechanisms.

Appendix A. Centralized transportation planning model

After collaboration, all transportation requests of a carrier alliance are reassigned to its members and the vehicle routes for serv-

Table 2
Results of the random instances from 1 to 10.

Instance	1	2	3	4	5	6	7	8	9	10
Pre-collaboration profits	111.7	69.1	166.3	309.6	119	135.5	137.3	187.6	166.4	258.3
	321.4	44.4	152.1	129.9	123.6	100	117.2	156.8	117.7	73.9
	132.3	180	137.8	235.4	99.5	283.4	330.8	142	287.9	113.9
Post-collaboration total profit	674	381.8	562.7	867.8	436.6	671.5	731.3	593.6	710.3	531
Profit allocation of mechanism 1	141.183	92.55	214.717	399.2	163.127	196.825	171.283	221.367	222.533	300.4
	361.833	65.6	170.167	187.95	148.616	168.175	172.383	213.167	148.283	81.8
	170.984	223.65	177.816	280.65	124.767	306.5	387.634	159.066	339.484	148.8
Profit allocation of mechanism 2	138.816	90.3358	218.545	400.612	165.263	199.22	166.218	221.064	224.699	300.231
	363.325	62.5532	165	187.602	147.573	173.087	174.632	216.343	144.934	82.05
	171.859	228.911	179.155	279.586	123.764	299.193	390.451	156.193	340.667	148.719
Profit allocation of mechanism 3	145.908	85.502	212.155	352.484	145.284	172.341	156.738	236.489	186.017	280.926
	366.092	70.9	172	224.479	164.9	192.659	202.6	195.464	152.42	87.66
	162	225.398	178.545	290.837	126.416	306.5	371.962	161.647	371.863	162.414

Table 3
Results of the random instances from 11 to 20.

Instance	11	12	13	14	15	16	17	18	19	20
Pre-collaboration profits	172.9	48.8	155.3	263.2	162.7	146.4	13.9	228.5	127.4	74.9
	294.7	84.4	152.1	206.5	98.5	103.6	117.1	166.5	87.6	299.8
	124.4	189.6	285.4	218	33.6	110.2	248.5	318.2	92	356.6
Post-collaboration total profit	761.6	411.7	717.4	900.8	342.9	640.6	493.6	890	475.9	911.4
Profit allocation of mechanism 1	239.767	78.266	182.7	340.767	173.3	271.9	50.533	275.325	158.6	131.6
	352.117	115.517	197.7	254.466	119.1	186.6	171.833	199.4	184.3	356.1
	169.716	217.917	337	305.567	50.5	182.1	271.234	415.275	133	423.7
Profit allocation of mechanism 2	242.032	78.242	182.405	341.483	172.8	288.223	49.99	268.411	151.628	131.526
	352.31	115.733	197.785	251.938	119.52	181.267	178.314	190.417	200.433	356.017
	167.258	217.725	337.21	307.379	50.58	171.11	265.296	431.172	123.839	423.856
Profit allocation of mechanism 3	279.359	63.223	171.898	370.11	179.071	249.3	52.708	283.4	158.6	95.744
	335.815	134.238	205.585	276.394	125.275	206.5	168.192	199.4	184.3	400.993
	146.426	214.239	339.917	254.296	38.554	184.8	272.7	407.2	133	414.663

ing the requests assigned to each carrier are determined based on a centralized collaborative transportation planning model. Firstly, the notations used in the model are given as follows.

Indices	
i, j, n	node index, $i, j, n = 1, \ldots, N$, where N represents the number of nodes in the transportation network
m	carrier index, $m = 1, \ldots, M$, where M represents the number of carriers
k	vehicle index, $k \in V, V = V_1 \cup V_2 \cdots \cup V_m$, where V_m represents the set of vehicles owned by carrier m
l	request index, $l = 1, \ldots, L$, where L represents the number of requests owned by all carriers in the transportation network, Each request is associated with a pickup/delivery task from one node to another node
Parameters	
CS	the carrier set, which includes all carriers in the alliance
S_i	the set of requests with pickup node i
E_i	the set of requests with delivery node i
u_l	quantity delivered on request l
p_l	ask price for request l (the price paid by a shipper to a carrier for serving the request)
C_m	vehicle capacity for each vehicle of carrier m
o	the set of depot nodes, $o = \{o_1, o_2, \cdots, o_m\}, o \subseteq \{1, \ldots, N\}$, o_m is the depot node of carrier m
c_{ij}^m	transportation cost from node i to j for each vehicle of carrier m, where $c_{ij}^m = c_{ji}^m$ and the triangle inequality, $c_{in}^m + c_{nj}^m \geqslant c_{ij}^m$, holds for any i, j, n and m with $n \neq i$, $n \neq j$, and $c_{ii}^m = 0$
τ_{ij}^k	the travelling time from node i to node j for each vehicle k
a_i	the earliest pickup/delivery time at node i
b_i	the latest pickup/delivery time at node i
T_{ij}	a large number, $T_{ij} = b_j - a_i$
Variables	
y_{lm}	1 if request l is allocated to carrier m, 0 otherwise
q_{ij}^k	quantity of product transported through arc (i, j) by vehicle k
x_{ij}^k	number of times that arc (i, j) is visited by vehicle k
t_i^k	the time at which vehicle k leaves from node i

Note that, without loss of generality, we assume that each node is associated with at most one request in the following model. The node is either the pickup location or the delivery location of a single request. If some nodes are associated with multiple requests, the model is still applicable if we replace each of the nodes by a number of duplicate nodes such that each duplicate node is associated with a single request, and set the travelling time and the transportation cost between any pair of the duplicate nodes to

zero. With this assumption, t_i^k is well defined since only one vehicle visits each node i for performing its associated pickup/delivery task.

With the notation, the centralized collaborative transportation planning model, which is a mixed integer programming model and denoted by *CTP*, is given as follows.

Model *CTP*:

$$Z^{CTP} = Max\left(\sum_{m=1}^{M}\sum_{l=1}^{L} p_l \cdot y_{lm} - \sum_{m=1}^{M}\sum_{k\in V_m}\sum_{i=1}^{N}\sum_{j=1,j\neq i}^{N} c_{ij}^m \cdot x_{ij}^k\right) \tag{19}$$

Subject to:

$$\sum_{j=1,j\neq i}^{N} x_{ij}^k = \sum_{j=1,j\neq i}^{N} x_{ji}^k, \quad i = 1,\ldots,N, \quad i \neq o_m, \quad k \in V_m, \quad m = 1,\ldots,M, \tag{20}$$

$$\sum_{j=1}^{N} x_{o_m j}^k = \sum_{j=1}^{N} x_{j o_m}^k = 1, \quad k \in V_m, \quad m = 1,\ldots,M, \tag{21}$$

$$q_{ij}^k \leqslant C_m \cdot x_{ij}^k, \quad i,j = 1,\ldots,N, \quad i \neq j, \quad k \in V_m, \quad m = 1,\ldots,M, \tag{22}$$

$$\sum_{k\in V_m}\sum_{j=1,j\neq i}^{N} q_{ij}^k - \sum_{k\in V_m}\sum_{j=1,j\neq i}^{N} q_{ji}^k = \sum_{l\in P_i} d_l \cdot y_{lm} - \sum_{l\in D_i} d_l \cdot y_{lm}, \quad i = 1,\ldots,N, \quad i \neq \{o_m\}, \quad m = 1,\ldots,M, \tag{23}$$

$$\sum_{j=1,j\neq o_m}^{N} q_{o_m j}^k = \sum_{l\in P_{o_m}} d_l \cdot y_{lk} - \sum_{l\in D_{o_m}} d_l \cdot y_{lm}, k \in V_m, \quad m = 1,\ldots,M, \tag{24}$$

$$\sum_{j=1,j\neq o_m}^{N} q_{j o_m}^k = 0, \quad k \in V_m, \quad m = 1,\ldots,M, \tag{25}$$

$$\sum_{m=1}^{M} y_{lm} \leqslant 1, l = 1,\ldots,L, \tag{26}$$

$$\sum_{k\in V_m}\sum_{j=1,j\neq i}^{N} x_{ij}^k \leqslant 1, \quad i = 1,\ldots,N, \quad i \notin o_m, \quad m = 1,\ldots,M, \tag{27}$$

$$t_j^k \geqslant t_i^k + \tau_{ij}^k \cdot x_{ij}^k - T_{ij} \cdot (1 - x_{ij}^k), i, \quad j = 1,\ldots,N, \quad j \neq o_m, \quad i \neq j, \quad k \in V_m, \quad m = 1,\ldots,M, \tag{28}$$

$$0 \leqslant a_i \leqslant t_i^k \leqslant b_i, \quad i = 1,\ldots,N, \quad k \in V, \tag{29}$$

$$\sum_{j=1,j\neq i}^{N} x_{ij}^k \leqslant y_{lm}, \quad i = 1,\ldots,N, \quad i \neq \{o_m\}, \quad l \in P_i \cup D_i, k \in V_m, \quad m = 1,\ldots,M, \tag{30}$$

$$x_{ij}^k \geqslant 0, x_{ij}^k \in Z, \quad i,j = 1,\ldots,N, \quad i \neq j, k \in V \tag{31}$$

$$y_{lm} \in \{0,1\}, y_{lm} \in Z, \quad l = 1,\ldots,L, \quad m = 1,\ldots,M, \tag{32}$$

$$q_{ij}^k \geqslant 0, q_{ij}^k \in R, \quad i,j = 1,\ldots,N, \quad i \neq j, \quad k \in V, \tag{33}$$

The objective function (19) represents the post-collaboration total profit of the carrier alliance. Constraints (20) ensure that the number of times of vehicle k leaving from a node (excluding depot nodes) is equal to the number of times of vehicle k arriving at the node. Constraints (21) guarantee that each vehicle of each carrier starts with and ends at its depot node. When $\sum_{j=1,j\neq o_m}^{N} x_{o_m j}^k = 1,\ k \in V_m, m = 1,\ldots,M$, vehicle k is used; otherwise if $x_{o_m o_m}^k = 1,\ k \in V_m, m = 1,\ldots,M$, vehicle k is not used. Constraints (22) are vehicle capacity constraints. Constraints (23) are flow conservation equations, assuring the flow balance at each node (except for all depot nodes) for each carrier. Constraints (24) and (25) are flow conservation equations on the depot node of each carrier, which ensure that only empty vehicle is returned to its depot node. Constraints (26) guarantee that each request is assigned to at most one carrier. Constraints (27) ensure that each customer node is visited at most once by vehicles of each carrier. Constraints (28) indicate the relationship between the times at which any two nodes are visited by a vehicle, which also ensure that every vehicle starts with and ends at its depot node. Constraints (29) are time window constraints for pickup and delivery operations of all requests. Since at most one pickup or delivery operation is associated with each node, each node is associated with a single time window constraint. Constraints (30) tighten the model *CTP*, which imply that a node is visited by a vehicle of carrier k only if it is the pickup or delivery location node of a request assigned to the carrier.

The above model can also be used for calculating the pre-collaboration profit/cost of each carrier. In order to do so, we set the number of carriers (M) to 1 in model *CTP*, which means that only one carrier is in the transportation network, then *CTP* becomes the transportation planning model for a single carrier before collaboration, based on which the pre-collaboration profit and cost of the carrier can be calculated. Furthermore, if for any given coalition *CSS*, $CSS \subseteq CS$, the post-collaboration total profit and total cost of *CSS* can also be calculated by model *CTP* with only the carriers in *CSS* considered in the model.

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